

# HARRY HERNÁNDEZ ARIAS

## Software Engineer

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### PROFESSIONAL SUMMARY

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Software Engineer with fullstack experience in the government and private sectors, building cross-platform systems, system integrations, APIs/backend services and data analytics modules. I work with AWS, Firebase, Python (FastAPI), SQL (PostgreSQL/MySQL) and the JavaScript/TypeScript ecosystem in React, Node.js and NestJS. I have focused my profile on LLM-based systems, prompt design, RAG architectures and agents, built on AWS Bedrock and Grok.

### PROFESSIONAL EXPERIENCE

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#### Tribunal de Justicia Administrativa de Tabasco

Software Engineer *May 2026 – Present*

- Adapted and integrated the Mexico City online court filing system for the Tabasco Court, normalizing the data model and the document catalogs across both institutions, which automated the judicial workflow of 5 unitary chambers.
- Built a centralized analytics and reporting module with optimized SQL queries on PostgreSQL, delivering unified operational tracking across all chambers and reducing the time spent producing Excel reports.

#### XpectreLabs — YogoApp freelance project

Backend Developer / Fullstack | *December 2025 – July 2026*

- Designed and built the complete backend of a ride-hailing app in NestJS, TypeScript and Firestore. Modular REST API, authentication and roles, driver verification, paid memberships with Pub/Sub, Firebase integration and push notifications.
- Deployed and operated the infrastructure on AWS Lightsail, with storage and static hosting of the landing page on S3; responsible for deployments and error monitoring (Sentry).
- Supported the development of the driver and passenger mobile apps, built with React Native and Firebase.

#### Secretaría de Gubernatura del Estado de Tabasco

Fullstack Developer *December 2025 – April 2026*

- Developed a government indicator management and monitoring platform with dashboards and visualizations, consolidating data from 9 agencies and giving visibility into strategic indicators. Built with TypeScript, Angular, Express.js and PostgreSQL.
- Translated the requirements of non-technical agencies into functional specifications and scalable data models, reducing rework caused by ambiguity.

#### Productos de Computo Gil (PCGIL)

Fullstack Developer *August 2024 – November 2025*

- Led technical support, maintenance and the rollout of updates for the SclarSoft and SclarSoft Black mobile apps, supporting active users and reducing the mean time to resolution of incidents.
- Developed a cross-platform livestock management application with React, React Native and Expo, delivering 11 modules including frontend, backend and integration tests.
- Integrated third-party services, authentication flows and RESTful APIs into two Node.js/Express backends.

#### Productos de Computo Gil (PCGIL)

Frontend Developer (Internship) *May 2024 – August 2024*

- Designed responsive, high-performance interfaces in Figma for an internal application, and built reusable, responsive components for web frontends with React.

## EDUCATION AND CERTIFICATIONS

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**Universidad Tecnológica de Tabasco** — *September 2022 – April 2026*

Engineering Degree in Cross-Platform Software Development and Management

**Anthropic Claude Partner Network** — May 2026 - June 2026

Courses: Agent Skills · Building with the Claude API · Model Context Protocol · Claude Code in Action

**Samsung Innovation Campus · Universidad de Monterrey** — October 2023 – March 2024

Internet of Things (IoT) and Artificial Intelligence with Python

**Oracle Next Education · Alura Latam** — March 2023 – September 2023

Backend and data analysis

## ACHIEVEMENTS AND PROJECTS

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**NASA International Space Apps Challenge 2025** | Global Nominee 2025

Selected as a Global Nominee among more than 1,290 nominated teams worldwide, after developing a system to make complex air quality data easier to understand; using NASA Earth observation data and other sources to deliver clear visualizations, forecasts and health recommendations for Mexico City and Los Angeles.

**AI agent development with AWS Bedrock and Grok** | Google Cloud Run and Lablab.ai hackathons

Built AI agents for trip planning and fake news detection on AWS Bedrock and Grok under hackathon time constraints, designing system prompts, structured JSON outputs and input/output guardrails.

**Organizer and lead of the AWS User Group Cárdenas** (Tabasco, Mexico).

## TECHNICAL SKILLS

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**Languages:** Python, SQL, JavaScript, TypeScript

**Data:** PostgreSQL, MySQL, Firebase

**Backend:** FastAPI, Node.js, Express.js, NestJS

**Frontend:** React, Next.js, React Native, Expo, Tailwind CSS

**AI tools and AI-assisted development:** Claude Code, Cursor, GitHub, Docker, AWS